

GAUTAM BUDDHA UNIVERSITY



SCHOOL OF ICT

DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING

DETAILED SYLLABI (SEMESTER III-IV)

M.TECH. (COMPUTER SCIENCE AND ENGINEERING) SPECIALIZATION IN DATA SCIENCE

ACADEMIC BATCH: 2025-2027

SEMESTER III

Signature

Cloud and Edge Computing			
Course Code:	CS621	Course Credits:	3
Course Category:	PCC	Course (U / P):	P
Course Year (U / P):	2P	Course Semester (U / P):	3P
No. of Lectures + Tutorials (Hrs/Week):	03 + 00	Mid Sem. Exam Hours:	1.5
Total No. of Lectures (L + T):	45 + 00	End Sem. Exam Hours:	3
COURSE OBJECTIVES			
1. Explain the principles and terminology of the foundations of Cloud and Edge Computing.			
2. Analyse problems and trade-offs concerning cloud architecture and virtualization.			
3. Apply suitable methods drawn from distributed data and compute.			
4. Design solutions that integrate edge-cloud continuum.			
5. Evaluate approaches and responsible practice relating to security, reliability and operations.			
COURSE OUTCOMES			
At the end of the course, the students should be able to:			
1 Explain the concepts and terminology of the foundations of Cloud and Edge Computing.			
2. Analyse problems using the methods covered under cloud architecture and virtualization.			
3. Apply appropriate techniques from distributed data and compute to realistic datasets or systems.			
4. Design a solution that integrates concepts from the edge-cloud continuum.			
5. Evaluate alternative approaches using technical, performance and reliability criteria.			
6. Develop and communicate a responsible, reproducible solution drawing on security, reliability and operations.			

UNIT I: FOUNDATIONS OF DISTRIBUTED COMPUTING

Evolution from clusters and grids to cloud and edge computing; Definition and essential characteristics of cloud computing — on-demand self-service, broad network access, resource pooling, rapid elasticity and measured service; Overview of service models (IaaS, PaaS, SaaS) and deployment models (public, private, hybrid, multi-cloud); Drivers for edge computing — latency, bandwidth, privacy and data locality; The cloud-to-edge continuum at a glance; Representative cloud and edge use cases; Benefits, limitations and cost/performance trade-offs..

UNIT II: CLOUD ARCHITECTURE AND VIRTUALIZATION

IaaS, PaaS and SaaS; Virtual machines and containers; Orchestration; Service discovery; Autoscaling; Multi-tenancy; Cloud storage; Hypervisor types; Full and para-virtualization; Hardware support; Namespaces and control groups; Container images and registries; VM-container comparison; Orchestration control loops; Resource isolation and scheduling; Public, private, hybrid and multi-cloud models; Regions and availability zones; Reference architectures; Elasticity policies; Resource provisioning; Load balancing; Cloud service selection; Performance, availability and cost comparison; Worked examples and case analysis.

UNIT III: DISTRIBUTED DATA AND COMPUTE

Distributed file systems; Key-value and NoSQL stores; Partitioning and replication; MapReduce, stream processing and serverless workflows; Data locality; Shuffling and partition strategies; Distributed joins; Lineage and recomputation; Batch-stream unification; Data skew; Checkpointing; Back-pressure; Execution-plan analysis; Scalability experiments; Single-writer and multi-writer replication; Leader-follower and leaderless designs; Quorum reads and writes; Conflict-free replicated data types; Consistency anomalies; Isolation levels; Compensating actions; Failure recovery; Worked examples and case analysis; Comparative evaluation of techniques.

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UNIT IV: EDGE-CLOUD CONTINUUM

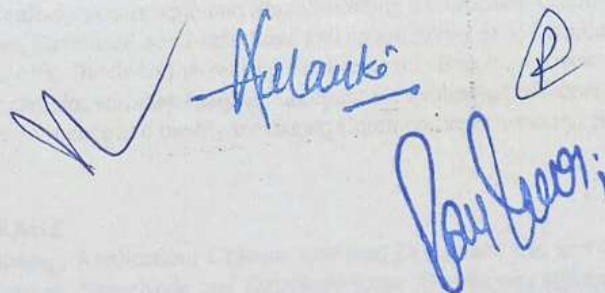
Fog and edge architectures; Placement and offloading; Latency, bandwidth and energy trade-offs; IoT integration; Federated edge learning; Public, private, hybrid and multi-cloud models; Regions and availability zones; Reference architectures; Elasticity policies; Resource provisioning; Load balancing; Cloud service selection; Performance, availability and cost comparison; Edge node taxonomy; Cloudlets and gateways; Task-placement formulations; Computation offloading; Mobility support; Intermittent connectivity; Edge caching; Privacy-preserving analytics; Latency-energy modelling; Smart-city and industrial-IoT examples; Worked examples and case analysis.

UNIT V: SECURITY, RELIABILITY AND OPERATIONS

Zero-trust basics; Identity and access; Observability; SRE indicators; Resilience patterns; Cost governance; Sustainable computing; Case studies; Availability and reliability measures; Redundancy; Health checks; Graceful degradation; Bulkheads; Recovery-point and recovery-time objectives; Backup and disaster recovery; Fault injection; Chaos experiments; Reliability-cost trade-offs; Logs, metrics and traces; Service-level indicators, objectives and error budgets; Capacity planning; Deployment strategies; Runbooks; Incident response; Root-cause analysis; Post-incident reviews; Operational dashboards; Worked examples and case analysis.

Text / Reference books

1. M. van Steen and A. S. Tanenbaum, Distributed Systems, 4th ed., distributed-systems.net, 2023.
2. T. Erl, R. Puttini and Z. Mahmood, Cloud Computing: Concepts, Technology & Architecture, Pearson.
3. T. Hwang, G. Fox and J. Dongarra, Distributed and Cloud Computing, Morgan Kaufmann.
4. R. Buyya and A. V. Dastjerdi (eds.), Internet of Things: Principles and Paradigms, Morgan Kaufmann.
5. C. Pahl et al., Cloud Container Technologies: a State-of-the-Art Review, IEEE Transactions on Cloud Computing.



Big Data Analytics			
Course Code:	CD623	Course Credits:	3
Course Category:	PCC	Course (U / P):	P
Course Year (U / P):	2P	Course Semester (U / P):	3P
No. of Lectures + Tutorials (Hrs/Week):	03 + 00	Mid Sem. Exam Hours:	1.5
Total No. of Lectures (L + T):	45	End Sem. Exam Hours:	3
COURSE OBJECTIVES			
1. Explain the principles and terminology of big data foundations.			
2. Analyse problems and trade-offs concerning distributed storage.			
3. Apply suitable methods drawn from batch processing.			
4. Design solutions that integrate streaming analytics.			
5. Evaluate approaches and responsible practice relating to production analytics.			
COURSE OUTCOMES			
At the end of the course, the students should be able to:			
1. Explain the fundamental concepts and terminology of big data foundations.			
2. Analyse problems using the methods covered under distributed storage.			
3. Apply appropriate techniques from batch processing to realistic datasets or systems.			
4. Design a solution that integrates concepts from streaming analytics.			
5. Evaluate alternative approaches using technical, performance and reliability criteria.			
6. Develop and communicate a responsible, reproducible solution drawing on production analytics.			

UNIT I: BIG DATA FOUNDATIONS

Data characteristics; Distributed architectures; Storage and compute trade-offs; Data lifecycle and governance; Five Vs of big data; Scale-up versus scale-out; Shared-nothing architecture; Commodity clusters; Workload characterization; Structured, semi-structured and unstructured data; Serialization formats; Distributed-system trade-offs; Batch and streaming requirements; Benchmark selection; Total cost of ownership; Worked examples and case analysis; Comparative evaluation of techniques; Implementation considerations; Performance and quality measures; Common limitations and failure scenarios.

UNIT II: DISTRIBUTED STORAGE

HDFS and object storage; Partitioning; Replication; Column families; Document and key-value stores; Consistency; HDFS architecture; NameNode and DataNode roles; Blocks and replication; Object-storage concepts; Partition keys; Consistent hashing; Document, key-value and wide-column stores; CAP trade-offs; Read and write paths; Compaction; Secondary indexes; Erasure coding; Storage-format and compression comparison; Worked examples and case analysis; Comparative evaluation of techniques; Implementation considerations; Performance and quality measures; Common limitations and failure scenarios.

UNIT III: BATCH PROCESSING

MapReduce; DAG execution; Spark RDD/DataFrame model; Joins; Optimization; Resource management; MapReduce programming model; Mapper, combiner, partitioner and reducer; Shuffle and sort; Data locality; Speculative execution; DAG scheduling; Spark RDDs and DataFrames; Lazy evaluation; Joins and aggregations; Caching; Skew mitigation; Execution-plan analysis; Resource and performance tuning; Worked examples and case analysis; Comparative evaluation of techniques; Implementation considerations; Performance and quality measures; Common limitations and failure scenarios; Contemporary tools and practices.

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UNIT IV: STREAMING ANALYTICS

Event time; Windows; Watermarks; State; Delivery semantics; Kafka and stream-processing architectures; Event streams; Producers, consumers and brokers; Kafka topics and partitions; Offsets and consumer groups; Event time and processing time; Tumbling, sliding and session windows; Late data; Stateful processing; Checkpointing; Back-pressure; Throughput and latency measurement; Worked examples and case analysis; Comparative evaluation of techniques; Implementation considerations; Performance and quality measures; Common limitations and failure scenarios; Contemporary tools and practices.

UNIT V: PRODUCTION ANALYTICS

Lakehouse and warehouse patterns; Orchestration Lakehouse;; Security; Cost; Responsible analytics; Case studies; Bronze-silver-gold layers; Incremental processing; Schema evolution; Lineage; Data-quality rules; Access control; Privacy; Cost governance; Service-level objectives; Incident recovery; Sustainable analytics; Worked examples and case analysis; Comparative evaluation of techniques; Implementation considerations; Performance and quality measures; Common limitations and failure scenarios; Contemporary tools and practices; Scalability considerations; Reproducibility practices.

Text/Reference books:

1. J. Leskovec, A. Rajaraman and J. Ullman, Mining of Massive Datasets, 3rd ed., Cambridge.
2. M. Kleppmann, Designing Data-Intensive Applications, O'Reilly.
3. B. Chambers and M. Zaharia, Spark: The Definitive Guide, O'Reilly.
4. T. White, Hadoop: The Definitive Guide, 4th ed., O'Reilly.
5. J. Kreps et al., Kafka: a Distributed Messaging System for Log Processing.

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Big Data Analytics Lab			
Course Code:	CD681	Course Credits:	2
Course Category:	PCC	Course (U / P):	P
Course Year (U / P):	2P	Course Semester (U / P):	3P
No. of Labs (Hrs/Week):	1 (4 HRS)	Mid Sem. Exam Hours:	—
Total No. of Labs:	10	End Sem. Exam Hours:	3
COURSE OBJECTIVES			
1. Configure and use contemporary tools required for the laboratory domain.			
2. Implement core algorithms and system components through reproducible experiments.			
3. Measure correctness, performance, robustness and resource trade-offs.			
4. Integrate multiple components into a working domain-specific solution.			
5. Document, demonstrate and critically evaluate responsible engineering practice.			
COURSE OUTCOMES			
At the end of the course, the students should be able to:			
1. Set up and operate the required laboratory platforms and development environment.			
2. Implement and test core techniques using appropriate datasets, simulators or systems.			
3. Collect and analyse experimental evidence using suitable metrics.			
4. Diagnose failures and improve a solution through systematic experimentation.			
5. Develop an integrated mini-project with reproducible code and documentation.			

PRACTICALS

Note: Each practical shall include objective, design/algorithm, implementation, test data, observations, results and a brief critical analysis. Faculty may adapt platforms while retaining the stated learning outcomes.

Lab 1: Distributed Environment and Data Ingestion

1. Configure a local/cluster big-data environment.
2. Ingest structured and semi-structured data.
3. Profile data volume, schema and quality.

Lab 2: Distributed Storage

1. Store and partition data in HDFS/object storage.
2. Compare text, Parquet and ORC formats.
3. Measure compression, scan cost and partition pruning.

Lab 3: MapReduce Processing

1. Implement word count and inverted index.
2. Design a join or aggregation job.
3. Analyze shuffle, skew and combiner effectiveness.

Lab 4: Spark DataFrames and SQL

1. Build transformations and analytical queries.
2. Inspect logical and physical plans.
3. Optimize caching, partitioning and joins.

Lab 5: Distributed Machine Learning

1. Train a scalable classification or clustering pipeline.
2. Tune and evaluate with reproducible splits.
3. Compare single-node and distributed execution.

Lab 6: NoSQL Data Modelling

1. Model data in a document or wide-column store.
2. Implement access patterns and indexes.
3. Evaluate consistency and performance trade-offs.

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Lab 7: Streaming with Kafka

1. Produce and consume events with schema management.
2. Handle partitions, offsets and failures.
3. Measure throughput and end-to-end latency.

Lab 8: Stateful Stream Analytics

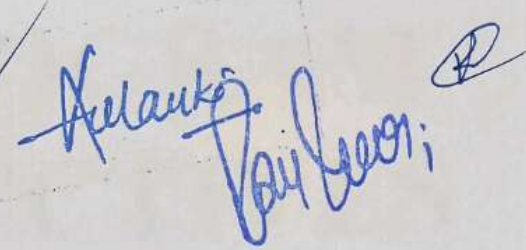
1. Implement windows, watermarks and state.
2. Handle late data and delivery semantics.
3. Build a real-time anomaly or trend detector.

Lab 9: Lakehouse Pipeline

1. Build bronze-silver-gold transformations.
2. Implement schema evolution, quality tests and lineage.
3. Orchestrate incremental and idempotent runs.

Lab 10: Integrated Analytics Project

1. Develop an end-to-end batch-plus-stream pipeline.
2. Add monitoring, access control and cost/performance analysis.
3. Present reproducible analytics and limitations.

PROGRAM ELECTIVES

Prof. Dr. S. S. S. S.

Data Mining and Knowledge Discovery			
Course Code:	CD631	Course Credits:	
Course Category:	PEC	Course (U / P):	P
Course Year (U / P):	2P	Course Semester (U / P):	3P
No. of Lectures + Tutorials (Hrs/Week):	03 + 00	Mid Sem. Exam Hours:	1.5
Total No. of Lectures (L + T):	45	End Sem. Exam Hours:	3
COURSE OBJECTIVES			
1. Explain the principles and terminology of the knowledge discovery process.			
2. Analyse problems and trade-offs concerning frequent patterns.			
3. Apply suitable methods drawn from classification.			
4. Design solutions that integrate clustering and anomaly detection.			
5. Evaluate approaches and responsible practice relating to advanced mining.			
COURSE OUTCOMES			
At the end of the course, the students should be able to:			
1. Explain the fundamental concepts and terminology of the knowledge discovery process.			
2. Analyse problems using the methods covered under frequent patterns.			
3. Apply appropriate techniques from classification to realistic datasets or systems.			
4. Design a solution that integrates concepts from clustering and anomaly detection.			
5. Evaluate alternative approaches using technical, performance and reliability criteria.			
6. Develop and communicate a responsible, reproducible solution drawing on advanced mining.			

UNIT I: KNOWLEDGE DISCOVERY PROCESS

Data understanding; Preprocessing; Missing data; Feature construction; Similarity; Evaluation design; Business and analytical objectives; CRISP-DM/KDD stages; Data audit; Cleaning and integration; Transformation; Feature selection; Validation protocol; Pattern interpretation; Deployment and feedback; Edge node taxonomy; Cloudlets and gateways; Task-placement formulations; Computation offloading; Mobility support; Intermittent connectivity; Edge caching; Privacy-preserving analytics; Latency-energy modelling; Smart-city and industrial-IoT examples; Worked examples and case analysis; Comparative evaluation of techniques; Implementation considerations; Performance and quality measures.

UNIT II: FREQUENT PATTERNS

Association rules; Apriori; FP-growth; Sequential and graph patterns; Interestingness; Transaction representation; Support, confidence, lift and leverage; Apriori candidate generation; Downward-closure property; FP-tree construction; Closed and maximal itemsets; Sequential patterns; Association-rule pruning; Scalability; Redundant-rule filtering; Market-basket interpretation; Worked examples and case analysis; Comparative evaluation of techniques; Implementation considerations; Performance and quality measures; Common limitations and failure scenarios; Contemporary tools and practices; Scalability considerations; Reproducibility practices; Ethical and privacy implications.

UNIT III: CLASSIFICATION

Trees; Rules; Nearest neighbours; Ensembles; Imbalance; Calibration; Evaluation; Logistic regression and decision-tree baselines; K-nearest neighbours; Naive Bayes; Support-vector machines; Bagging; Random forests; Boosting; Multiclass strategies; Class imbalance; Probability calibration; ROC and precision-recall analysis; Cost-sensitive evaluation; Error analysis; Worked examples and case analysis; Comparative evaluation of techniques; Implementation considerations; Performance and quality measures; Common limitations and failure scenarios; Contemporary tools and practices; Scalability considerations; Reproducibility practices.

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UNIT IV: CLUSTERING AND ANOMALY DETECTION

Partitioning, hierarchical and density methods; Cluster validation; Outlier detection; Distance and similarity measures; K-means and k-medoids; Hierarchical linkage; DBSCAN; Spectral clustering; Cluster-number selection; Silhouette and stability measures; High-dimensional clustering; Density and isolation-based outlier detection; Novelty detection; Threshold selection; Anomaly-score evaluation; Worked examples and case analysis; Comparative evaluation of techniques; Implementation considerations; Performance and quality measures; Common limitations and failure scenarios; Contemporary tools and practices; Scalability considerations.

UNIT V: ADVANCED MINING

Text, web and graph mining; Recommender systems; Scalable mining; Privacy, fairness and deployment; Text and web mining; Sequence and graph mining; Topic modelling; Concept drift; Streaming algorithms; Distributed pattern mining; Privacy-preserving mining; Explainability; Pattern validation; Deployment feedback; Domain case studies; Worked examples and case analysis; Comparative evaluation of techniques; Implementation considerations; Performance and quality measures; Common limitations and failure scenarios; Contemporary tools and practices; Scalability considerations.

Text/Reference books

1. J. Han, M. Kamber and J. Pei, Data Mining: Concepts and Techniques, 3rd ed., Morgan Kaufmann.
2. P.-N. Tan, M. Steinbach, A. Karpatne and V. Kumar, Introduction to Data Mining, 2nd ed., Pearson.
3. J. Leskovec, A. Rajaraman and J. Ullman, Mining of Massive Datasets, 3rd ed., Cambridge.
4. I. H. Witten et al., Data Mining, 4th ed., Morgan Kaufmann.
5. C. C. Aggarwal, Outlier Analysis, 2nd ed., Springer.

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Time Series Analysis and Forecasting			
Course Code:	CD633	Course Credits:	3
Course Category:	PEC	Course (U / P):	P
Course Year (U / P):	2P	Course Semester (U / P):	3P
No. of Lectures + Tutorials (Hrs/Week):	03 + 00	Mid Sem. Exam Hours:	1.5
Total No. of Lectures (L + T):	45 S	End Sem. Exam Hours:	3
COURSE OBJECTIVES			
1. Explain the principles and terminology of time-series foundations.			
2. Analyse problems and trade-offs concerning classical models.			
3. Apply suitable methods drawn from multivariate and state-space models.			
4. Design solutions that integrate modern forecasting.			
5. Evaluate approaches and responsible practice relating to forecast evaluation.			
COURSE OUTCOMES			
At the end of the course, the students should be able to:			
1. Explain the fundamental concepts and terminology of time-series foundations.			
2. Analyse problems using the methods covered under classical models.			
3. Apply appropriate techniques from multivariate and state-space models to realistic datasets or systems.			
4. Design a solution that integrates concepts from modern forecasting.			
5. Evaluate alternative approaches using technical, performance and reliability criteria.			
6. Develop and communicate a responsible, reproducible solution drawing on forecast evaluation.			

UNIT I: TIME-SERIES FOUNDATIONS

Stationarity; Autocorrelation; Transformations; Decomposition; Exploratory analysis; Temporal indexing and resampling; Trend, cycle and seasonality; Lag plots; Autocorrelation and partial autocorrelation; Missing observations; Outliers; Leakage-safe temporal splitting; Baseline forecasting; Exploratory visualization; Forecast horizons; Worked examples and case analysis; Comparative evaluation of techniques; Implementation considerations; Performance and quality measures; Common limitations and failure scenarios; Contemporary tools and practices; Scalability considerations; Reproducibility practices; Ethical and privacy implications; Tool-supported exercises.

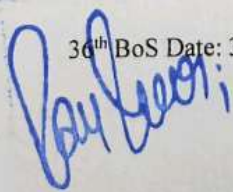
UNIT II: CLASSICAL MODELS

AR, MA, ARMA and ARIMA; Estimation; Diagnostics; Seasonal models; Exponential smoothing; Autoregressive, moving-average and ARMA processes; ARIMA identification; Parameter estimation; AIC and BIC; Residual diagnostics; Seasonal ARIMA; Simple and Holt-Winters exponential smoothing; Dynamic regression; Intervention analysis; Multi-step forecasting; Worked examples and case analysis; Comparative evaluation of techniques; Implementation considerations; Performance and quality measures; Common limitations and failure scenarios; Contemporary tools and practices; Scalability considerations; Reproducibility practices.

UNIT III: MULTIVARIATE AND STATE-SPACE MODELS

VAR; Cointegration; Kalman filtering; Dynamic regression; Interventions; Cross-correlation; Vector autoregression; Granger causality; Error-correction models; Latent-state formulation; Kalman filtering and smoothing; Dynamic linear models; Parameter estimation; Diagnostics; Multivariate forecasting; Uncertainty intervals; Worked examples and case analysis; Comparative evaluation of techniques; Implementation considerations; Performance and quality measures; Common limitations and failure scenarios; Contemporary tools and practices; Scalability

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considerations; Reproducibility practices; Ethical and privacy implications; Tool-supported exercises; Industry case studies.

UNIT IV: MODERN FORECASTING

Feature-based ML; Tree ensembles; Neural sequence models; Temporal convolution and transformers; Lag and rolling features; Tree-based forecasting; Recursive, direct and multi-output strategies; Global versus local models; RNN and LSTM; Temporal convolution; Transformer-based forecasting; Ensembling; Hyperparameter tuning; Computational and interpretability trade-offs; Worked examples and case analysis; Comparative evaluation of techniques; Implementation considerations; Performance and quality measures; Common limitations and failure scenarios; Contemporary tools and practices; Scalability considerations.

UNIT V: FORECAST EVALUATION

Rolling-origin validation; Probabilistic forecasts; Hierarchical series; Anomalies; Monitoring and case studies; Expanding and sliding windows; MAE, RMSE, MAPE and scaled errors; Prediction intervals; Quantile loss; Probabilistic calibration; Hierarchical reconciliation; Intermittent demand; Anomaly detection; Drift monitoring; Forecast-value assessment; Worked examples and case analysis; Comparative evaluation of techniques; Implementation considerations; Performance and quality measures; Common limitations and failure scenarios; Contemporary tools and practices; Scalability considerations; Reproducibility practices.

Text / Reference books:

1. R. J. Hyndman and G. Athanasopoulos, Forecasting: Principles and Practice, 3rd ed., OTexts.
2. R. H. Shumway and D. S. Stoffer, Time Series Analysis and Its Applications, 4th ed., Springer.
3. P. J. Brockwell and R. A. Davis, Introduction to Time Series and Forecasting, 3rd ed., Springer.
4. J. D. Hamilton, Time Series Analysis, Princeton.
5. C. Chatfield, The Analysis of Time Series, 6th ed., CRC.

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Agentie AI			
Course Code:	CD635	Course Credits:	3
Course Category:	PEC	Course (U / P):	P
Course Year (U / P):	2P	Course Semester (U / P):	3P
No. of Lectures + Tutorials (Hrs/Week):	03 + 00	Mid Sem. Exam Hours:	1.5
Total No. of Lectures (L + T):	45	End Sem. Exam Hours:	3

COURSE OBJECTIVES

1. Explain the principles and terminology of agentic artificial intelligence and autonomous intelligent agents.
2. Analyse problems and trade-offs concerning planning, reasoning and decision-making in agentic systems.
3. Apply suitable methods drawn from large language models, tool integration and multi-agent collaboration.
4. Design agentic AI systems for solving real-world autonomous tasks.
5. Evaluate approaches and responsible practice relating to autonomous AI agents.

COURSE OUTCOMES

At the end of the course, the students should be able to:

1. Explain the fundamental concepts and terminology of agentic AI and intelligent agents.
2. Analyse problems using the methods covered under planning, reasoning and autonomous decision-making.
3. Apply appropriate techniques for tool usage, workflow automation and multi-agent collaboration.
4. Design an agentic AI solution for a specified real-world application.
5. Evaluate alternative agentic AI approaches using technical, performance and reliability criteria.
6. Develop and communicate a responsible, reproducible agentic AI solution.

UNIT I: FOUNDATIONS OF AGENTIC AI

Intelligent agents; Agentic AI fundamentals; Characteristics of autonomous agents; Agent architectures; Perception, reasoning and action; Reactive and deliberative agents; Goal-oriented behaviour; Planning concepts; Memory and context management; Agent life cycle; Applications; Worked examples and case analysis; Comparative evaluation of techniques; Implementation considerations; Performance measures; Common limitations and failure scenarios; Contemporary tools and practices; Safety and robustness considerations.

UNIT II: AGENT REASONING AND TOOL INTEGRATION

Reasoning strategies; Task decomposition; Planning and execution; Tool calling; API integration; External knowledge retrieval; Workflow automation; Retrieval-augmented generation (RAG); Memory management; Agent orchestration; Worked examples and case analysis; Comparative evaluation of techniques; Implementation considerations; Performance measures; Common limitations and failure scenarios; Contemporary tools and practices; Safety and robustness considerations; Simulation-based exercises.

UNIT III: MULTI-AGENT SYSTEMS

Multi-agent systems; Collaborative agents; Communication protocols; Coordination strategies; Task allocation; Negotiation; Conflict resolution; Distributed problem solving; Human-agent collaboration; Agent evaluation; Worked examples and case analysis; Comparative evaluation of techniques; Implementation considerations; Performance measures; Common limitations and failure scenarios; Contemporary tools and practices; Safety and robustness considerations; Reproducibility

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UNIT IV: AGENTIC AI FRAMEWORKS AND APPLICATIONS

Agent development frameworks; Workflow design; Conversational agents; Coding assistants; Autonomous research assistants; Enterprise automation; Robotics applications; Software engineering applications; Decision support systems; Healthcare and finance use cases; Worked examples and case analysis; Comparative evaluation of techniques; Implementation considerations; Performance measures; Common limitations and failure scenarios; Contemporary tools and practices; Safety and robustness considerations.

UNIT V: DEPLOYMENT, EVALUATION AND RESPONSIBLE AGENTIC AI

Agent evaluation; Benchmarking; Reliability; Scalability; Monitoring; Explainability; Human oversight; Fairness; Bias; Privacy; Security; Ethical and legal considerations; Governance; Responsible AI; Emerging trends in Agentic AI; Worked examples and case analysis; Comparative evaluation of techniques; Implementation considerations; Performance measures; Common limitations and failure scenarios; Future research directions.

Text/Reference books:

1. Stuart Russell and Peter Norvig, Artificial Intelligence: A Modern Approach, 4th Edition, Pearson.
2. Wooldridge, M., An Introduction to MultiAgent Systems, 2nd Edition, Wiley.
3. Bratman, M., Intentions, Plans, and Practical Reason, Harvard University Press.
4. Foster, D., Generative Deep Learning, 2nd Edition, O'Reilly Media.
5. Luger, G. F., Artificial Intelligence: Structures and Strategies for Complex Problem Solving, 8th Edition,

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 - A signature that appears to read "Vijay Kumar".
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Data Engineering and Modern Data Warehousing			
Course Code:	CD637	Course Credits:	3
Course Category:	PEC	Course (U / P):	P
Course Year (U / P):	2P	Course Semester (U / P):	3P
No. of Lectures + Tutorials (Hrs/Week):	03 + 00	Mid Sem. Exam Hours:	1.5
Total No. of Lectures (L + T):	45	End Sem. Exam Hours:	3
COURSE OBJECTIVES			
1. Explain the principles and terminology of data architecture.			
2. Analyse problems and trade-offs concerning ingestion and integration.			
3. Apply suitable methods drawn from transformation and orchestration.			
4. Design solutions that integrate warehousing and performance.			
5. Evaluate approaches and responsible practice relating to dataops and governance.			
COURSE OUTCOMES			
At the end of the course, the students should be able to:			
1. Explain the fundamental concepts and terminology of data architecture.			
2. Analyse problems using the methods covered under ingestion and integration.			
3. Apply appropriate techniques from transformation and orchestration to realistic datasets or systems.			
4. Design a solution that integrates concepts from warehousing and performance.			
5. Evaluate alternative approaches using technical, performance and reliability criteria.			
6. Develop and communicate a responsible, reproducible solution drawing on dataops and governance.			

UNIT I: DATA ARCHITECTURE

Operational and analytical systems; Dimensional modelling; Data vault; Lake, warehouse, Operational and analytical workloads; Source-system analysis; Star and snowflake schemas; Fact and dimension tables; Slowly changing dimensions; Data lake, warehouse; Data mesh and data fabric concepts; Architecture decision criteria; Worked examples and case analysis; Comparative evaluation of techniques; Implementation considerations; Performance and quality measures; Common limitations and failure scenarios; Contemporary tools and practices.

UNIT II: INGESTION AND INTEGRATION

Batch and CDC pipelines; APIs; Messaging; Data contracts; Metadata; Full and incremental ingestion; Log-based and query-based change-data capture; Message queues; Connectors; Schema registry; Ordering; Deduplication; Replay; Error queues; Idempotent loading; Metadata and provenance; Worked examples and case analysis; Comparative evaluation of techniques; Implementation considerations; Performance and quality measures; Common limitations and failure scenarios; Contemporary tools and practices; Scalability considerations; Reproducibility practices; Ethical and privacy implications.

UNIT III: TRANSFORMATION AND ORCHESTRATION

ELT; Distributed SQL; Workflow DAGs; Idempotency; Backfills; Testing and CI/CD; ETL and ELT; SQL and distributed transformations; Scheduling and triggers; Dependency management; Retries; Incremental models; Unit and integration tests; Environment promotion; Lineage; Failure recovery; Worked examples and case analysis; Comparative evaluation of techniques; Implementation considerations; Performance and quality measures; Common limitations and failure scenarios; Contemporary tools and practices; Scalability considerations; Reproducibility practices; Ethical and privacy implications.

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UNIT IV: WAREHOUSING AND PERFORMANCE

Columnar storage; Partitioning; Clustering; Materialization; Query optimization; Semantic layers; Dimensional modelling; Surrogate keys; Partitioning and clustering; Indexes; Materialized views; Distributed SQL; Query plans; Join strategies; Workload management; Caching; Benchmark design; Cost and performance optimization; Worked examples and case analysis; Comparative evaluation of techniques; Implementation considerations; Performance and quality measures; Common limitations and failure scenarios; Contemporary tools and practices; Scalability considerations; Reproducibility practices; Ethical and privacy implications.

UNIT V: DATAOPS AND GOVERNANCE

Lineage; Catalogues; Quality; Access control; Privacy; Cost and reliability; Data ownership and stewardship; Business glossary; Data-quality dimensions and rules; SLAs; Role-based access; Masking; Retention; Audit; Incident management; Compliance and FinOps; Worked examples and case analysis; Comparative evaluation of techniques; Implementation considerations; Performance and quality measures; Common limitations and failure scenarios; Contemporary tools and practices; Scalability considerations; Reproducibility practices; Ethical and privacy implications; Tool-supported exercises.

Text/Reference books:

1. M. Kleppmann, Designing Data-Intensive Applications, O'Reilly.
2. J. Reis and M. Housley, Fundamentals of Data Engineering, O'Reilly.
3. R. Kimball and M. Ross, The Data Warehouse Toolkit, 3rd ed., Wiley.
4. M. Fowler, Patterns of Enterprise Application Architecture, Addison-Wesley.
5. D. Wells, Data Quality: The Accuracy Dimension, Morgan Kaufmann.

Digital Transformation Using AI			
Course Code:	CD639	Course Credits:	3
Course Category:	PEC	Course (U / P):	P
Course Year (U / P):	2P	Course Semester (U / P):	3P
No. of Lectures + Tutorials (Hrs/Week):	03 + 00	Mid Sem. Exam Hours:	1.5
Total No. of Lectures (L + T):	45	End Sem. Exam Hours:	3

COURSE OBJECTIVES

1. Explain the concepts, strategies, and technologies of digital transformation using Artificial Intelligence.
2. Analyse AI-driven business processes and enterprise digital transformation frameworks.
3. Apply AI, data analytics, and automation techniques to solve business problems.
4. Design intelligent digital transformation solutions using cloud, AI, and emerging technologies.
5. Evaluate AI-enabled digital transformation initiatives considering business value, ethics, security, and sustainability.

COURSE OUTCOMES

At the end of the course, the students should be able to:

1. Explain the principles and technologies of AI-enabled digital transformation.
2. Analyse enterprise processes suitable for AI-driven transformation.
3. Apply AI techniques to automate and optimize business operations.
4. Design intelligent digital transformation solutions for real-world applications.
5. Evaluate digital transformation initiatives using business, technical, and ethical considerations.
6. Assess the principles and methodologies of AI-enabled digital transformation.

UNIT I: DIGITAL TRANSFORMATION FOUNDATIONS

Digital transformation concepts; Industry 4.0; Digital business models; Enterprise digital transformation; AI as a transformation driver; Digital maturity models; Business process transformation; Digital ecosystems; Cloud computing fundamentals; Data-driven organizations; Intelligent enterprises; Digital strategy; Customer-centric transformation; Innovation management; Emerging technologies; Enterprise case studies; Comparative evaluation of approaches; Performance measures; Contemporary tools and practices.

UNIT II: AI FOR BUSINESS PROCESS TRANSFORMATION

Business process automation; Robotic Process Automation (RPA); Intelligent automation; Machine learning for enterprises; Predictive analytics; Process mining; Workflow optimization; Recommendation systems; Customer analytics; AI-powered decision support; Chatbots and virtual assistants; Document intelligence; OCR; NLP for enterprises; Enterprise knowledge management; AI-enabled CRM and ERP; Comparative evaluation of techniques; Implementation considerations; Industry applications.

UNIT III: GENERATIVE AI AND ENTERPRISE INTELLIGENCE

Generative AI; Foundation Models; Large Language Models (LLMs); Prompt Engineering; Retrieval-Augmented Generation (RAG); Enterprise search; AI Copilots; AI Agents; Knowledge graphs; Semantic search; Document summarization; Intelligent report generation; Code generation; Workflow automation; Multimodal AI; AI-assisted decision making; Function calling; Enterprise AI platforms; Worked examples and case analysis; Contemporary AI tools and practices.

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UNIT IV: DIGITAL PLATFORMS, CLOUD AND DATA ANALYTICS

Cloud-native applications; Data engineering; Data lakes; Data warehouses; Big Data analytics; IoT integration; Edge AI; Digital twins; Business Intelligence; Data visualization; Real-time analytics; MLOps; AI model deployment; API integration; Enterprise integration; Cybersecurity for digital platforms; Scalability; Performance optimization; Industry case studies.

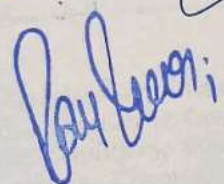
UNIT V: GOVERNANCE, SECURITY AND FUTURE OF DIGITAL TRANSFORMATION

AI governance; Responsible AI; Explainable AI; AI ethics; Privacy and security; Data governance; Regulatory compliance; Risk management; AI adoption strategies; Change management; Digital leadership; AI project management; Sustainability; Green AI; Enterprise architecture; ROI measurement; Digital transformation frameworks; Future trends; Case studies; Comparative evaluation of techniques; Contemporary tools and practices.

Text / Reference books :

1. M. A. Hernán and J. M. Robins, Causal Inference: What If, Chapman & Hall/CRC.
2. J. Pearl, M. Glymour and N. Jewell, Causal Inference in Statistics, 2nd ed., Wiley.
3. S. Cunningham, Causal Inference: The Mixtape, Yale.
4. G. W. Imbens and D. B. Rubin, Causal Inference, Cambridge.
5. R. Kohavi, D. Tang and Y. Xu, Trustworthy Online Controlled Experiments, Cambridge.

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Causal Inference and Experimentation			
Course Code:	CD641	Course Credits:	3
Course Category:	PEC	Course (U / P):	P
Course Year (U / P):	2P	Course Semester (U / P):	3P
No. of Lectures + Tutorials (Hrs/Week):	03 + 00	Mid Sem. Exam Hours:	1.5
Total No. of Lectures (L + T):	45	End Sem. Exam Hours:	3
COURSE OBJECTIVES			
1. Explain the principles and terminology of causal foundations.			
2. Analyse problems and trade-offs concerning randomized experiments.			
3. Apply suitable methods drawn from observational methods.			
4. Design solutions that integrate quasi-experimental designs.			
5. Evaluate approaches and responsible practice relating to heterogeneous and scalable causality.			
COURSE OUTCOMES			
At the end of the course, the students should be able to:			
1. Explain the fundamental concepts and terminology of causal foundations.			
2. Analyse problems using the methods covered under randomized experiments.			
3. Apply appropriate techniques from observational methods to realistic datasets or systems.			
4. Design a solution that integrates concepts from quasi-experimental designs.			
5. Evaluate alternative approaches using technical, performance and reliability criteria.			
6. Develop and communicate a responsible, reproducible solution drawing on heterogeneous and scalable causality.			

UNIT I: CAUSAL FOUNDATIONS

Potential outcomes; DAGs; Identification; Confounding; Selection bias; Estimands; Treatment, outcome and covariates; Average treatment effects; Counterfactual reasoning; Causal DAGs; Back-door and front-door criteria; Collider and selection bias; Consistency; Positivity; SUTVA; Identification assumptions; Worked examples and case analysis; Comparative evaluation of techniques; Implementation considerations; Performance and quality measures; Common limitations and failure scenarios; Contemporary tools and practices; Scalability considerations; Reproducibility practices; Ethical and privacy implications; Tool-supported exercises.

UNIT II: RANDOMIZED EXPERIMENTS

Design; Blocking; Cluster and factorial experiments; Power; Noncompliance; Sequential testing; Randomization unit; Control and treatment groups; Blocking and stratification; Cluster and factorial designs; Sample size; Statistical power; Minimum detectable effect; Guardrail metrics; Attrition; Novelty and carry-over effects; Worked examples and case analysis; Comparative evaluation of techniques; Implementation considerations; Performance and quality measures; Common limitations and failure scenarios; Contemporary tools and practices; Scalability considerations; Reproducibility practices.

UNIT III: OBSERVATIONAL METHODS

Matching; Propensity scores; Weighting; Regression adjustment; Sensitivity analysis; Propensity-score estimation; Matching and calipers; Inverse-probability weighting; Covariate balance; Overlap diagnostics; Doubly robust estimation; Selection bias; Missing data; Sensitivity to unmeasured confounding; Rosenbaum analysis; Worked examples and case analysis; Comparative evaluation of techniques; Implementation considerations; Performance and quality measures; Common limitations and failure scenarios; Contemporary tools and practices; Scalability considerations; Reproducibility practices; Ethical and privacy implications; Tool-supported exercises.

UNIT IV: QUASI-EXPERIMENTAL DESIGNS

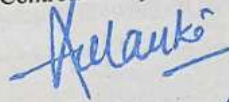
Instrumental variables; Regression discontinuity; Difference-in-differences; Synthetic controls; Relevance and exclusion assumptions; Bandwidth and continuity checks; Parallel-trends assessment; Event studies; Placebo and robustness tests; Interpretation limitations; Worked examples and case analysis; Comparative evaluation of techniques; Implementation considerations; Performance and quality measures; Common limitations and failure scenarios; Contemporary tools and practices; Scalability considerations; Reproducibility practices; Ethical and privacy implications; Tool-supported exercises; Industry case studies; Recent research directions.

UNIT V: HETEROGENEOUS AND SCALABLE CAUSALITY

Causal ML; Uplift; Mediation; Interference; Online experiments; Ethics and reporting; Conditional treatment effects; Causal forests; Uplift modelling; Meta-learners; Network experiments; Online experimentation platforms; Variance reduction; Multiple testing; Scalable estimation; Responsible decision making; Worked examples and case analysis; Comparative evaluation of techniques; Implementation considerations; Performance and quality measures; Common limitations and failure scenarios; Contemporary tools and practices; Scalability considerations; Reproducibility practices; Ethical and privacy implications; Tool-supported exercises.

Text/Reference books

1. M. A. Hernán and J. M. Robins, Causal Inference: What If, Chapman & Hall/CRC.
2. J. Pearl, M. Glymour and N. Jewell, Causal Inference in Statistics, 2nd ed., Wiley.
3. S. Cunningham, Causal Inference: The Mixtape, Yale.
4. G. W. Imbens and D. B. Rubin, Causal Inference, Cambridge.
5. R. Kohavi, D. Tang and Y. Xu, Trustworthy Online Controlled Experiments, Cambridge.



Graph Machine Learning and Network Analytics			
Course Code:	CD643	Course Credits:	3
Course Category:	PEC	Course (U / P):	P
Course Year (U / P):	2P	Course Semester (U / P):	3P
No. of Lectures + Tutorials (Hrs/Week):	03 + 00	Mid Sem. Exam Hours:	1.5
Total No. of Lectures (L + T):	45	End Sem. Exam Hours:	3

COURSE OBJECTIVES

1. Explain the principles and terminology of graph foundations.
2. Analyse problems and trade-offs concerning network analytics.
3. Apply suitable methods drawn from graph representation learning.
4. Design solutions that integrate graph neural networks.
5. Evaluate approaches and responsible practice relating to advanced applications.

COURSE OUTCOMES

At the end of the course, the students should be able to:

1. Explain the fundamental concepts and terminology of graph foundations.
2. Analyse problems using the methods covered under network analytics.
3. Apply appropriate techniques from graph representation learning to realistic datasets or systems.
4. Design a solution that integrates concepts from graph neural networks.
5. Evaluate alternative approaches using technical, performance and reliability criteria.
6. Develop and communicate a responsible, reproducible solution drawing on advanced applications.

UNIT I: GRAPH FOUNDATIONS

Graph representations; Paths and connectivity; Centrality; Motifs; Random graph models; Adjacency and incidence representations; Degree distributions; Paths, walks and connectivity; Components; Clustering coefficient; Centrality measures; Assortativity; Random-graph models; Small-world and scale-free properties; Null-model comparison; Worked examples and case analysis; Comparative evaluation of techniques; Implementation considerations; Performance and quality measures; Common limitations and failure scenarios; Contemporary tools and practices; Scalability considerations; Reproducibility practices; Ethical and privacy implications.

UNIT II: NETWORK ANALYTICS

Community detection; Link prediction; Diffusion; Influence; Temporal and heterogeneous networks; Modularity; Louvain and spectral methods; Similarity-based link prediction; Diffusion and cascade models; Influence maximization; Temporal networks; Multiplex and heterogeneous graphs; Network visualization; Evaluation measures; Worked examples and case analysis; Comparative evaluation of techniques; Implementation considerations; Performance and quality measures; Common limitations and failure scenarios; Contemporary tools and practices; Scalability considerations; Reproducibility practices; Ethical and privacy implications.

UNIT III: GRAPH REPRESENTATION LEARNING

Matrix factorization; Random-walk embeddings; Node2vec; Graph kernels; DeepWalk; Random-walk sampling; Negative sampling; Proximity-preserving objectives; Structural and attributed embeddings; Transductive and inductive learning; Node classification; Link-prediction evaluation; Embedding visualization; Worked examples and case analysis; Comparative evaluation of techniques; Implementation considerations; Performance and quality measures; Common limitations and failure scenarios; Contemporary tools and practices; Scalability considerations; Reproducibility practices; Ethical and privacy implications; Tool-supported exercises; Industry case studies.

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UNIT IV: GRAPH NEURAL NETWORKS

Message passing; GCN, GraphSAGE and attention; Graph-level learning; Scalability; Message-passing framework; Neighbourhood aggregation; GCN; GraphSAGE; Graph attention; Node, edge and graph-level tasks; Permutation invariance; Sampling and mini-batching; Over-smoothing; Over-squashing; Benchmark design and evaluation; Worked examples and case analysis; Comparative evaluation of techniques; Implementation considerations; Performance and quality measures; Common limitations and failure scenarios; Contemporary tools and practices; Scalability considerations; Reproducibility practices; Ethical and privacy implications.

UNIT V: ADVANCED APPLICATIONS

Knowledge graphs; Recommendation; Fraud and scientific graphs; Explainability; Fairness; Evaluation; Knowledge-graph completion; Fraud and anomaly detection; Molecular and scientific graphs; Traffic forecasting; Social-network analysis; Temporal graphs; Heterogeneous information networks; Explanation; Privacy; Application-specific evaluation; Worked examples and case analysis; Comparative evaluation of techniques; Implementation considerations; Performance and quality measures; Common limitations and failure scenarios; Contemporary tools and practices; Scalability considerations; Reproducibility practices; Ethical and privacy implications.

Text/Reference books

1. W. L. Hamilton, Graph Representation Learning, Morgan & Claypool.
2. J. Leskovec, A. Rajaraman and J. Ullman, Mining of Massive Datasets, 3rd ed., Cambridge.
3. M. Newman, Networks, 2nd ed., Oxford.
4. A. Barabási, Network Science, Cambridge.
5. Z. Wu et al., A Comprehensive Survey on Graph Neural Networks, IEEE TNNLS.

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Responsible and Explainable AI			
Course Code:	CD645	Course Credits:	3
Course Category:	PEC	Course (U / P):	P
Course Year (U / P):	2P	Course Semester (U / P):	3P
No. of Lectures + Tutorials (Hrs/Week):	03 + 00	Mid Sem. Exam Hours:	1.5
Total No. of Lectures (L + T):	45	End Sem. Exam Hours:	3

COURSE OBJECTIVES

1. Explain the principles and terminology of responsible AI foundations.
2. Analyse problems and trade-offs concerning fairness.
3. Apply suitable methods drawn from explainability.
4. Design solutions that integrate privacy, robustness and safety.
5. Evaluate approaches and responsible practice relating to assurance and governance.

COURSE OUTCOMES

At the end of the course, the students should be able to:

1. Explain the fundamental concepts and terminology of responsible AI foundations.
2. Analyse problems using the methods covered under fairness.
3. Apply appropriate techniques from explainability to realistic datasets or systems.
4. Design a solution that integrates concepts from privacy, robustness and safety.
5. Evaluate alternative approaches using technical, performance and reliability criteria.
6. Develop and communicate a responsible, reproducible solution drawing on assurance and governance.

UNIT I: RESPONSIBLE AI FOUNDATIONS

Socio-technical systems; Harms; Stakeholders; Accountability; Governance frameworks; Socio-technical system boundaries; Stakeholders and affected communities; Harm and risk taxonomies; Accountability roles; Responsible-AI principles; Lifecycle governance; Impact assessment; Documentation; Contestability; Human oversight; Standards and regulatory context; Worked examples and case analysis; Comparative evaluation of techniques; Implementation considerations; Performance and quality measures; Common limitations and failure scenarios; Contemporary tools and practices; Scalability considerations; Reproducibility practices; Ethical and privacy implications.

UNIT II: FAIRNESS

Sources of bias; Group and individual metrics; Impossibility results; Mitigation; Intersectionality; Protected attributes and proxies; Demographic parity; Equalized odds; Equality of opportunity; Predictive parity; Individual and intersectional fairness; Bias sources; Pre-, in- and post-processing mitigation; Utility-fairness trade-offs; Worked examples and case analysis; Comparative evaluation of techniques; Implementation considerations; Performance and quality measures; Common limitations and failure scenarios; Contemporary tools and practices; Scalability considerations; Reproducibility practices.

UNIT III: EXPLAINABILITY

Interpretable models; Local and global explanations; Feature attribution; Counterfactuals; Explanation evaluation; Intrinsic and post-hoc interpretation; Feature importance; Partial-dependence and ICE plots; LIME; SHAP; Prototypes; Counterfactual explanations; Faithfulness; Stability; Usefulness and human evaluation; Worked examples and case analysis; Comparative evaluation of techniques; Implementation considerations; Performance and quality measures; Common limitations and failure scenarios; Contemporary tools and practices; Scalability considerations; Reproducibility practices; Ethical and privacy implications; Tool-supported

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Exercises.

UNIT IV: PRIVACY, ROBUSTNESS AND SAFETY

Privacy attacks; Differential privacy; Adversarial robustness; Uncertainty; Human oversight; Data minimization; Anonymization limitations; Membership and model-inversion attacks; Privacy budget; Adversarial examples; Poisoning; Distribution shift; Robustness testing; Safe fallback; Human approval; Worked examples and case analysis; Comparative evaluation of techniques; Implementation considerations; Performance and quality measures; Common limitations and failure scenarios; Contemporary tools and practices; Scalability considerations; Reproducibility practices; Ethical and privacy implications; Tool-supported exercises; Industry case studies.

UNIT V: ASSURANCE AND GOVERNANCE

Documentation; Audits; Impact assessment; Model cards; Monitoring; Regulation and domain case studies; Model and data cards; Risk registers; Independent audit; Approval gates; Test evidence; Regulatory mapping; Access and change control; Incident escalation; Continuous monitoring; Redress; Model retirement; Governance metrics; Worked examples and case analysis; Comparative evaluation of techniques; Implementation considerations; Performance and quality measures; Common limitations and failure scenarios; Contemporary tools and practices; Scalability considerations.

Text / Reference books

1. S. Barocas, M. Hardt and A. Narayanan, Fairness and Machine Learning, fairmlbook.org.
2. C. Molnar, Interpretable Machine Learning, 2nd ed., 2022.
3. NIST, Artificial Intelligence Risk Management Framework 1.0.
4. E. Kamar, Directions in Hybrid Intelligence, IJCAI.
5. F. Doshi-Velez and B. Kim, Towards a Rigorous Science of Interpretable Machine Learning.

Handwritten signatures and initials:
A large blue checkmark, the signature "Aravind", a circled "P", and the signature "Bharathi".

Optimization Methods for Data Science			
Course Code:	CD647	Course Credits:	3
Course Category:	PEC	Course (U / P):	P
Course Year (U / P):	2P	Course Semester (U / P):	3P
No. of Lectures + Tutorials (Hrs/Week):	03 + 00	Mid Sem. Exam Hours:	1.5
Total No. of Lectures (L + T):	45	End Sem. Exam Hours:	3

COURSE OBJECTIVES

1. Explain the principles and terminology of convex analysis.
2. Analyse problems and trade-offs concerning first-order methods.
3. Apply suitable methods drawn from constrained optimization.
4. Design solutions that integrate stochastic and large-scale optimization.
5. Evaluate approaches and responsible practice relating to optimization in machine learning.

COURSE OUTCOMES

At the end of the course, the students should be able to:

1. Explain the fundamental concepts and terminology of convex analysis.
2. Analyse problems using the methods covered under first-order methods.
3. Apply appropriate techniques from constrained optimization to realistic datasets or systems.
4. Design a solution that integrates concepts from stochastic and large-scale optimization.
5. Evaluate alternative approaches using technical, performance and reliability criteria.
6. Develop and communicate a responsible, reproducible solution drawing on optimization in machine learning.

UNIT I: CONVEX ANALYSIS

Convex sets and functions; Norms; Projections; Smoothness; Optimality conditions; Affine sets; Epigraphs; Jensen's inequality; First- and second-order conditions; Lipschitz continuity; Strong convexity; Separating hyperplanes; Machine-learning modelling examples; Worked examples and case analysis; Comparative evaluation of techniques; Implementation considerations; Performance and quality measures; Common limitations and failure scenarios; Contemporary tools and practices; Scalability considerations; Reproducibility practices; Ethical and privacy implications; Tool-supported exercises; Industry case studies; Recent research directions.

UNIT II: FIRST-ORDER METHODS

Gradient and subgradient descent; Acceleration; Proximal methods; Coordinate descent; Step-size rules; Convergence rates; Momentum; Nesterov acceleration; Proximal gradient; Stopping criteria; Numerical stability; Sparse and composite objectives; Comparative convergence experiments; Worked examples and case analysis; Comparative evaluation of techniques; Implementation considerations; Performance and quality measures; Common limitations and failure scenarios; Contemporary tools and practices; Scalability considerations; Reproducibility practices; Ethical and privacy implications; Tool-supported exercises; Industry case studies.

UNIT III: CONSTRAINED OPTIMIZATION

Lagrange duality; KKT conditions; Penalty and barrier methods; Projected methods; Equality and inequality constraints; Feasible directions; Lagrangian; Weak and strong duality; Complementary slackness; Projected gradient; Augmented Lagrangian; Constrained modelling examples; Worked examples and case analysis; Comparative evaluation of techniques; Implementation considerations; Performance and quality measures; Common limitations and failure scenarios; Contemporary tools and practices; Scalability considerations; Reproducibility practices; Ethical and privacy implications; Tool-supported exercises; Industry case studies.

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UNIT IV: STOCHASTIC AND LARGE-SCALE OPTIMIZATION

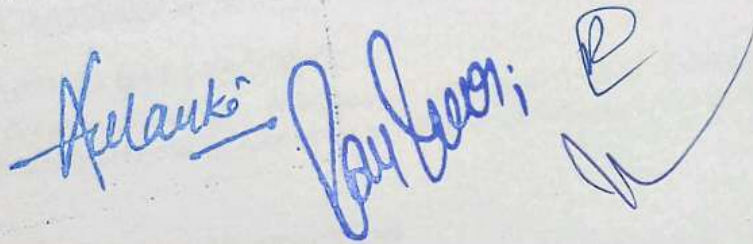
SGD; Variance reduction; Adaptive methods; Distributed and federated optimization; Stochastic and mini-batch gradients; Learning-rate schedules; Momentum; AdaGrad, RMSProp and Adam; Asynchronous updates; Communication efficiency; Convergence diagnostics; Scaling experiments; Worked examples and case analysis; Comparative evaluation of techniques; Implementation considerations; Performance and quality measures; Common limitations and failure scenarios; Contemporary tools and practices; Scalability considerations; Reproducibility practices; Ethical and privacy implications; Tool-supported exercises; Industry case studies.

UNIT V: OPTIMIZATION IN MACHINE LEARNING

Regularized regression; SVMs; Matrix factorisation; Sparse learning; Hyperparameter and non-convex optimization; Least-squares and logistic objectives; Support-vector machines; Sparse and group penalties; Hyperparameter optimization; Non-convex landscapes; Initialisation; Early stopping; Optimisation-generalisation relationship; Worked examples and case analysis; Comparative evaluation of techniques; Implementation considerations; Performance and quality measures; Common limitations and failure scenarios; Contemporary tools and practices; Scalability considerations; Reproducibility practices; Ethical and privacy implications; Tool-supported exercises; Industry case studies.

Text / Reference books

1. S. Boyd and L. Vandenberghe, Convex Optimization, Cambridge.
2. A. Beck, First-Order Methods in Optimization, SIAM.
3. D. P. Bertsekas, Nonlinear Programming, 3rd ed., Athena Scientific.
4. J. Nocedal and S. Wright, Numerical Optimization, 2nd ed., Springer.
5. S. Shalev-Shwartz and S. Ben-David, Understanding Machine Learning, Cambridge.



Data Security			
Course Code:	CD649	Course Credits:	3
Course Category:	PEC	Course (U / P):	P
Course Year (U / P):	2P	Course Semester (U / P):	3P
No. of Lectures + Tutorials (Hrs/Week):	03 + 00	Mid Sem. Exam Hours:	1.5
Total No. of Lectures (L + T):	45	End Sem. Exam Hours:	3

COURSE OBJECTIVES

1. To understand the fundamental concepts of security parameters, access control, System Design and Information Security.
2. To understand the fundamental concepts and working of various symmetric and asymmetric cryptography algorithms
3. To understand security threats, vulnerabilities, and protection mechanisms.
4. To understand the importance of Database security
5. To understand types of threats and also the importance of Firewall/Email/web security.

COURSE OUTCOMES

At the end of the course, the students should be able to:

1. Understand confidentiality, integrity, and availability along with security parameters
2. To understand various cryptographic algorithms.
3. Outline access control mechanisms and security policies.
4. Illustrate database security.
5. To understand the importance and implementation of Web security and Firewalls
6. To understand various cryptographic algorithms.

UNIT I: INTRODUCTION TO DATA SECURITY

The need for security, Security approaches, Principles of security CIA Triad (Confidentiality, Integrity, Availability): Security services, attacks, and mechanisms: Ethical, legal, and professional issues

UNIT II: CRYPTOGRAPHY & DATA PROTECTION

Introduction, plain text and ciphertext, substitution techniques, transposition techniques, encryption and decryption, symmetric and asymmetric key cryptography, steganography, key range and key size, possible types of attacks. Symmetric and asymmetric encryption: Hash functions and digital signatures: Classical and modern cyphers (DES, AES, RSA)

UNIT III: ACCESS CONTROL & PROGRAM SECURITY

Access Control Models: Discretionary, mandatory, and role-based access control (DAC, MAC, RBAC), control of access and information flow, Malicious code, viruses, worms, and Trojans: Operating system and memory protection

UNIT IV: DATABASE & INFORMATION SECURITY

Database & Information Security: Database vulnerabilities and SQL injection; Multilevel database security and inference control; Data anonymisation and privacy regulations. Special Topics: Data privacy, introduction to digital forensics, enterprise security specification

UNIT V: NETWORK & WEB SECURITY

Firewalls and intrusion detection systems (IDS); password management, Virus and related threats, Countermeasures, Firewall design principles, Types of firewalls; Email and web application security protocols; E-Mail Security: Pretty Good Privacy, S/MIME Web Security: Web security considerations, Secure Socket Layer and Transport Layer Security, Secure electronic transaction

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Intruders, Virus Risk management and organizational security policies Case Studies on Cryptography and security: Secure Inter Branch Payment Transactions, Cross site Scripting Vulnerability, Virtual Elections

Text/Reference books:

1. Cryptography and Network Security: William Stallings, Pearson Education, 4th Edition
2. Cryptography and Network Security: Atul Kahate, McGraw-Hill, 2nd Edition
3. Security Engineering, Ross Anderson.
4. Information Security: Principles and Practice, M. Stamp.
5. Cryptography and Network Security: C K Shyamala, N Harini, Dr T R Padmanabhan, Wiley India, 1st Edition.
6. Cryptography and Network Security: Forouzan Mukhopadhyay, McGraw Hill, 2nd Edition
7. Information Security, Principles and Practice: Mark Stamp, Wiley India.
8. Principles of Computer Security: WM.Arthur Conklin, Greg White, TMH
9. Introduction to Network Security: Neal Krawetz, CENGAGE Learning
6. Network 10. Security and Cryptography: Bernard Menezes, CENGAGE Learning
11. Security in Computing, C.P. Pfleeger, S.L. Pfleeger, J. Margulies.
12. Secure Programming HOWTO, David Wheeler.
13. Handbook of Database Security, M. Gertz, S. Jajodia.

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 - A signature that appears to read "P. Padmanabhan".
 - A signature that appears to read "C.P. Pfleeger".

SEMESTER IV

CD692: Dissertation Part-II

Prof. Dr. S. S. S. S.